Jiaming Chu

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chu-jiaming.github.io M.Eng. student, Shanghai Jiao Tong University

Education

present Sep 2025	School of Automation and Intelligent Sensing, **Shanghai Jiao Tong University** M.Eng. in Electronic Information
Jul 2025 Sep 2021	School of Instrumentation Science and Engineering, **Harbin Institute of Technology** B.Eng. in IntelliSense Engineering

Research

2026 **PoseGuard-NeRF: When Does Joint Camera-Pose Optimization Help NeRF?** page • code

- Controlled study of **45 vanilla-NeRF runs** (9 pose-noise levels × 5 methods) on when joint pose optimization helps or hurts.
- A camera learning-rate warm-up recovers most of the loss from noisy poses: 14.1 → **22.7 dB** PSNR at 5% translation noise.
- From perfect poses it drops PSNR 27.1 → 17.9 dB, yet a Sim(3) fit on training cameras recovers **+5.5 dB**: global drift, not a worse reconstruction.
- Released v1.0.0 with reproducible configs, GitHub Actions CI and uv environments.

Experience

Jul 2023 – Jan 2023 **Captain, Team Lilac 12** @ HIT Smartcar & Innovation Club

- Led vision support; built ROS packages, tuned navigation (SLAM, path planning) and wrote docs.

Apr 2024 – Nov 2023 **Team Member** @ HIT Center of Ultra-Precision Optoelectronic Instrument Engineering

- Single-beam 3-DoF ultra-precision laser interferometer: host communication and data visualization; the entry won a national second prize (see Awards).

Jan 2024 – Dec 2023 **Mathematical Modeling (Team Lead)**

- Built the models, wrote the code and the paper; won an ICM Honorable Mention.

Other Projects

Undergrad

- Qingzhou robot (C/C++, Python, Jetson Nano, STM32): SLAM, navigation, visual recognition.
- Vision-controlled car (Python, Raspberry Pi): drives by recognizing QR codes and hand gestures.
- Smart-home system (Python, Raspberry Pi): real-time sensing, logging, display and RF device control.

Honors & Awards

2024	**Honorable Mention**, Interdisciplinary Contest in Modeling (ICM), COMAP
2023	**Second National Prize**, 7th China (International) Sensor Innovation Contest
2023	**Provincial Second Prize**, National University Students Intelligent Car Race

Skills & Languages

Programming	Python, C/C++, MATLAB
Frameworks	PyTorch, OpenCV, NumPy
Systems	Linux, ROS
Dev Tools	Git, Conda, uv, LaTeX
Languages	Chinese (native), English (proficient)